ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Quadratic Equations** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 7	Sum $= -b/a$
2	**(a)** 9	$D = 0$: $4m^2 - 4k = 0$
3	**(d)** 14	Other root 7; product $= k$
4	**(b)** 3 and 5	Factorise
5	**(b)** 25	$D = b^2 - 4ac$
6	**(c)** two distinct real roots	$D = b^2 - 4ac = 12$
7	**(b)** 9	Sum $= -b/a$
8	**(d)** 16	$D = 0$: $4m^2 - 4k = 0$
9	**(c)** 24	Other root 8; product $= k$
10	**(a)** -4 and 6	Factorise

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Integers $x, x+1$: $x(x+1) = 72$ (1); $x^2 + x - 72 = 0$ (1).	1+1
12	Split the middle term and factorise (1); roots $\frac{3}{2}, 2$ (1).	1+1
13	$D = 0$: $(8)^2 - 4k = 0$ (1); $k = 16$ (1).	1+1
14	Integers $x, x+1$: $x(x+1) = 90$ (1); $x^2 + x - 90 = 0$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$D = (-7)^2 - 4(2)(3) = 25$ (1); $x = \frac{7 \pm 5}{4}$ (1); roots $x = 3, \frac{1}{2}$ (1).	1+1+1
16	$D = (-6)^2 - 4(1)(8) = 4$ (1); $x = \frac{6 \pm 2}{2}$ (1); roots $x = 4, 2$ (1).	1+1+1
17	$D = k^2 - 4(4) = 0$ (1); $k^2 = 16$ (1); $k = \pm 4$ (1).	1+1+1
	OR: $D = (-6)^2 - 4(1)(4) = 20 > 0$ (1); $x = \frac{6 \pm \sqrt{20}}{2}$ (1); roots $\frac{6+\sqrt{20}}{2}$ and $\frac{6-\sqrt{20}}{2}$ (1).	1+1+1
18	$D = (-2)^2 - 4(1)(-15) = 64$ (1); $x = \frac{2 \pm 8}{2}$ (1); roots $x = 5, -3$ (1).	1+1+1
	OR: $D = k^2 - 4(9) = 0$ (1); $k^2 = 36$ (1); $k = \pm 6$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Speed x: $\frac{360}{x} - \frac{360}{x+5} = 1$ (1); $1800 = x(x+5)$ so $x^2 + 5x - 1800 = 0$ (2); $(x-40)(x+45) = 0$ (1); speed 40 km/h (1).	1+2+1+1
	OR: Breadth x, length $x+7$: $x(x+7) = 330$ (1); $x^2 + 7x - 330 = 0$ (1); $(x-15)(x+22) = 0$ (1); $x = 15$ m (breadth), length 22 m (reject negative) (2).	1+1+1+2
20	Speed x: $\frac{420}{x} - \frac{420}{x+10} = 1$ (1); $4200 = x(x+10)$ so $x^2 + 10x - 4200 = 0$ (2); $(x-60)(x+70) = 0$ (1); speed 60 km/h (1).	1+2+1+1
	OR: Breadth x, length $x+9$: $x(x+9) = 220$ (1); $x^2 + 9x - 220 = 0$ (1); $(x-11)(x+20) = 0$ (1); $x = 11$ m (breadth), length 20 m (reject negative) (2).	1+1+1+2

Mistakes to avoid:

- Writing the formula as $\frac{-b \pm \sqrt{b^2+4ac}}{2a}$ or dividing only part of the numerator by $2a$.
- Forgetting the sign of b: for $x^2 - 5x + 6 = 0$, $-b = +5$.
- Not shifting everything to one side to get the standard form $ax^2 + bx + c = 0$ before using D or factorising.

- Cancelling a common factor containing x from both sides, which loses a root (e.g. $x(x-2)=3x$).
- Not rejecting a negative root for length, speed, age or time in word problems.
- Saying $D=0$ means no roots; it means two *equal* real roots.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Quadratic Equations** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** -56	$D = b^2 - 4ac$
2	**(d)** no real roots	$D = b^2 - 4ac = -8$
3	**(a)** 11	Sum $= -b/a$
4	**(c)** 25	$D = 0$: $4m^2 - 4k = 0$
5	**(a)** 36	Other root 9; product $= k$
6	**(b)** 2 and 9	Factorise
7	**(c)** 1	$D = b^2 - 4ac$
8	**(c)** two equal real roots	$D = b^2 - 4ac = 0$
9	**(a)** 8	Sum $= -b/a$
10	**(c)** 36	$D = 0$: $4m^2 - 4k = 0$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Split the middle term and factorise (1); roots $-\frac{1}{2}, 3$ (1).	1+1
12	$D = 0$: $(10)^2 - 4k = 0$ (1); $k = 25$ (1).	1+1
13	Integers $x, x+1$: $x(x+1) = 110$ (1); $x^2 + x - 110 = 0$ (1).	1+1
14	Split the middle term and factorise (1); roots $\frac{2}{3}, -1$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$D = (-5)^2 - 4(3)(1) = 13 > 0$ (1); $x = \frac{5 \pm \sqrt{13}}{6}$ (1); roots $\frac{5+\sqrt{13}}{6}$ and $\frac{5-\sqrt{13}}{6}$ (1).	1+1+1
16	$D = (-11)^2 - 4(2)(12) = 25$ (1); $x = \frac{11 \pm 5}{4}$ (1); roots $x = 4, \frac{3}{2}$ (1).	1+1+1
17	$D = k^2 - 4(16) = 0$ (1); $k^2 = 64$ (1); $k = \pm 8$ (1).	1+1+1
	OR: $D = (5)^2 - 4(2)(-3) = 49$ (1); $x = \frac{-5 \pm 7}{4}$ (1); roots $x = \frac{1}{2}, -3$ (1).	1+1+1
18	$D = (-4)^2 - 4(1)(-12) = 64$ (1); $x = \frac{4 \pm 8}{2}$ (1); roots $x = 6, -2$ (1).	1+1+1
	OR: $D = k^2 - 4(25) = 0$ (1); $k^2 = 100$ (1); $k = \pm 10$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Speed x: $\frac{240}{x} - \frac{240}{x+4} = 2$ (1); $960 = 2x(x+4)$ so $x^2 + 4x - 480 = 0$ (2); $(x-20)(x+24) = 0$ (1); speed 20 km/h (1).	1+2+1+1
	OR: Breadth x, length $x+5$: $x(x+5) = 500$ (1); $x^2 + 5x - 500 = 0$ (1); $(x-20)(x+25) = 0$ (1); $x = 20$ m (breadth), length 25 m (reject negative) (2).	1+1+1+2
20	Base x, altitude $x-7$: $x^2 + (x-7)^2 = 289$ (1); $2x^2 - 14x - 240 = 0$, i.e. $x^2 - 7x - 120 = 0$ (2); $(x-15)(x+8) = 0$ (1); base 15 cm, altitude 8 cm (1).	1+2+1+1
	OR: Ages x and $42 - x$: $(x-5)(37-x) = 175$ (1); simplifying $x^2 - 42x + 360 = 0$ (2); $(x-30)(x-12) = 0$ (1); ages 30 and 12 years (1).	1+2+1+1

Mistakes to avoid:

- Writing the formula as $\frac{-b \pm \sqrt{b^2+4ac}}{2a}$ or dividing only part of the numerator by $2a$.
- Forgetting the sign of b: for $x^2 - 5x + 6 = 0$, $-b = +5$.
- Not shifting everything to one side to get the standard form $ax^2 + bx + c = 0$ before using D or factorising.
- Cancelling a common factor containing x from both sides, which loses a root (e.g. $x(x-2) = 3x$).

- Not rejecting a negative root for length, speed, age or time in word problems.
- Saying $D = 0$ means no roots; it means two *equal* real roots.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Quadratic Equations** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 30	Other root 6; product $= k$
2	**(a)** -5 and 3	Factorise
3	**(c)** 0	$D = b^2 - 4ac$
4	**(d)** two equal real roots	$D = b^2 - 4ac = 0$
5	**(d)** 13	Sum $= -b/a$
6	**(b)** 49	$D = 0$: $4m^2 - 4k = 0$
7	**(a)** 60	Other root 10; product $= k$
8	**(a)** 1 and 10	Factorise
9	**(d)** 29	$D = b^2 - 4ac$
10	**(c)** no real roots	$D = b^2 - 4ac = -7$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$D = 0$: $(12)^2 - 4k = 0$ (1); $k = 36$ (1).	1+1
12	Integers $x, x+1$: $x(x+1) = 156$ (1); $x^2 + x - 156 = 0$ (1).	1+1
13	Split the middle term and factorise (1); roots $-\frac{5}{2}, 1$ (1).	1+1
14	$D = 0$: $(16)^2 - 4k = 0$ (1); $k = 64$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$D = (2)^2 - 4(1)(5) = -16 < 0$ (2); no real roots (1).	1+1+1
16	$D = (6)^2 - 4(1)(-7) = 64$ (1); $x = \dfrac{-6 \pm 8}{2}$ (1); roots $x = 1, -7$ (1).	1+1+1
17	$D = k^2 - 4(36) = 0$ (1); $k^2 = 144$ (1); $k = \pm 12$ (1).	1+1+1
	OR: $D = (-3)^2 - 4(5)(-2) = 49$ (1); $x = \dfrac{3 \pm 7}{10}$ (1); roots $x = 1, -\frac{2}{5}$ (1).	1+1+1
18	$D = (-5)^2 - 4(3)(-2) = 49$ (1); $x = \dfrac{5 \pm 7}{6}$ (1); roots $x = 2, -\frac{1}{3}$ (1).	1+1+1
	OR: $D = k^2 - 4(49) = 0$ (1); $k^2 = 196$ (1); $k = \pm 14$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Base x, altitude $x - 4$: $x^2 + (x-4)^2 = 400$ (1); $2x^2 - 8x - 384 = 0$, i.e. $x^2 - 4x - 192 = 0$ (2); $(x-16)(x+12) = 0$ (1); base 16 cm, altitude 12 cm (1).	1+2+1+1
	OR: Ages x and $36 - x$: $(x-6)(30-x) = 63$ (1); simplifying $x^2 - 36x + 243 = 0$ (2); $(x-27)(x-9) = 0$ (1); ages 27 and 9 years (1).	1+2+1+1
20	Base x, altitude $x - 3$: $x^2 + (x-3)^2 = 225$ (1); $2x^2 - 6x - 216 = 0$, i.e. $x^2 - 3x - 108 = 0$ (2); $(x-12)(x+9) = 0$ (1); base 12 cm, altitude 9 cm (1).	1+2+1+1
	OR: Ages x and $49 - x$: $(x-7)(42-x) = 196$ (1); simplifying $x^2 - 49x + 490 = 0$ (2); $(x-35)(x-14) = 0$ (1); ages 35 and 14 years (1).	1+2+1+1

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